



WEEK 1 · DAY 3

Why Traditional QA **Breaks**

Same question, different answers — and why that changes how we test.

Quality Thought · Gen AI Testing for Manual Testers

What we'll cover today



Non-determinism

Same input, different output



The oracle problem

You know 'good', not the exact answer



Why exact-match fails

`assertEquals` breaks down



What we check instead

Meaning, facts, format, safety

THE CORE PROBLEM

Same question → different answers



Like asking 3 friends

Ask three people the same thing and each phrases it differently — yet all are correct.

LLMs do the same.

For testers: a different wording is NOT automatically a bug.

Watch it work

SAME QUESTION → DIFFERENT ANSWERS (ALL CORRECT)

Q: “When will my order arrive?”

Animated — plays in slideshow mode

A KEY CONCEPT

The oracle problem



Normal software

You have an 'oracle' — the exact expected result.

Like checking a maths answer: $2 + 2 = 4$. Right or wrong.



Gen AI software

You know what a GOOD answer looks like, not the exact words.

Like grading an essay — judged on quality, not an exact string.

WHY EXACT-MATCH FAILS

Both answers are **correct** — but not equal



Expected (your script)

“Your order ships tomorrow.”

An exact-match check demands these EXACT words.



Actual (the model)

“It'll be on its way tomorrow!”

Same meaning, different words → exact-match wrongly FAILS it.

What we check instead of exact match



Contains key facts

Does it include the must-have details?



Means the same

Is it close in meaning to a good answer?



Right format

Valid JSON / required structure?



Stays safe

No leaks, no unsafe content?



Rubric score

A 1–5 quality score (later: LLM-as-judge)



Pass-rate over runs

Run N times — what % pass?

TESTING TIP

Use temperature to make tests **repeatable**



Pin it to 0

For regression tests, set temperature 0 so output is as stable as possible.

Fewer false failures from random wording.



But test reality too

Real users run higher temperatures.

So also test distributions: run many times, assert a pass-rate threshold.

LESSON OUTCOMES

What you can now do



Explain non-determinism with a simple analogy



Describe the oracle problem and why it matters



Explain why exact-match assertions fail on Gen AI



List what we check instead of an exact string

Check your understanding

Q1. True/False: a differently-worded answer is always a bug.

Q2. In Gen AI testing, do we have an exact 'oracle' for most answers?

Q3. Name two things we check instead of exact match.

Q4. What temperature do we use for repeatable regression tests?

QUICK CHECK

Answers



A1. False — same meaning in different words can still be correct.



A2. No — we usually know what 'good' looks like, not the exact text.



A3. Any two of: contains key facts, semantic similarity, format, safety, rubric score, pass-rate.



A4. 0 (the lowest) — to make output as repeatable as possible.

Your 30-minute lab + what's next



Today's mini-lab (30 min)

1. Pick one prompt (e.g. 'Summarise this paragraph: ...').
2. Send it 10 times at temperature 0; save the answers.
3. Send it 10 times at temperature 1; save the answers.
4. Note: how much did each set vary? Write 3 lines.
5. Save notes in your GitHub: week1/day3-notes.md



Coming up next

Day 4 — Designing Test Cases

Re-use your test-design skills (equivalence, boundary, negative) to write solid Gen AI test cases.